Panskura Banamali College (Autonomous)

B.Sc.(H) 4th Semester Examination -2021 (Online Mode)

Subject: Computer Science **Paper: CC8T**

Paper Name: Design and Analysis of Algorithms

F.M.: 40 **Time : 2 Hours**

Answer any four : **4x10=40**

1. Prove that if f1(n)=O(g1(n)) and f2(n)=O(g2(n)), then f1(n) + f2(n) = O(g1(n) + g2(n)). What is the time complexity of Merge sort? 8+2

2. What do you understand by algorithm evaluation? What is the significance of Big-oh, Omega and theta? 8+2

3. What is the basic principal of divide-and-conquer? What are the advantages of Merge sort over the quick sort algorithm? 8+2

4. Define spanning tree. Write Kruskal's algorithm for finding minimum cost spanning tree.Describe how Kruskal's algorithm is different from Prim's algorithm for finding minimum cost spanning tree. 2+ 6+ 2

5 . Extend the Dijkastra's algorithm to find All-pairs-shortest-path (APSP) problem. 10

6. Short note on : BFS , DFS. 5+5

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